API Documentation: Project Dalupotha

Base URL: https://api.dalupotha.factory.local/v1 **Content-Type:** application/json
Authentication: Bearer Token (JWT)

1. Global Error Codes

The Dalupotha API uses standard HTTP status codes to indicate the success or failure of an API request.

HTTP Status Code	Description	Example Scenario
200 OK	Success	Request was processed successfully.
201 Created	Created	A new resource (e.g., user, collection record) was created.
400 Bad Request	Validation Error	Missing required fields, invalid GPS coordinates.
401 Unauthorized	Auth Failure	Missing or invalid JWT token, invalid OTP.
403 Forbidden	Access Denied	User lacks the required role (e.g., TA trying to approve a machine rent).
404 Not Found	Resource Missing	Requested Supplier ID or Item ID does not exist.
409 Conflict	Data Conflict	Duplicate passbook number or contact number during registration.
500 Internal Error	Server Failure	Database connection failure or unhandled exception.

Standard Error Response Format:

```JSON
{
  "error": "Bad Request",
  "message": "Invalid GPS coordinates provided.",
  "code": 400,
  "timestamp": "2026-02-22T08:30:00Z"
}
```

2. Authentication & Registration Endpoints

2.1 Staff & Transport Agent Login

Authenticates internal users and returns a JWT for subsequent requests.

- **Endpoint:** POST /auth/login
- **Access:** Public

Request Body:

```JSON
{
  "username": "TA_Kumara",
  "password": "securepassword123"
}
```

Response (200 OK):

```JSON
{
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI...",
  "user": {
    "user_id": "a1b2c3d4-e5f6-7890-abcd-1234567890ab",
    "role": "TA",
    "username": "TA_Kumara"
  },
  "expires_in": 3600
}
```

2.2 Small Holder Registration (OTP Flow)

Registers a new Small Holder with GPS mapping.

- **Endpoint:** POST /auth/small-holder/register

- **Access:** Public

Request Body:

```JSON
{
  "full_name": "Saman Perera",
  "contact_number": "0771234567",
  "passbook_no": "PB-88992",
  "gps_lat": 6.8412,
  "gps_long": 79.9015,
  "in_charge_no": "EXT-004",
  "otp_code": "481516"
}
```

Response (201 Created):

```JSON
{
  "message": "Registration successful",
  "supplier_id": "c9d8e7f6-a5b4-3210-fedc-0987654321fe"
}
```

3. Field Collection & Logistics Endpoints

3.1 Sync Leaf Collections (Transport Agent)

Used by the Transport Agent mobile app to sync offline-captured green leaf data to the central database. Supports batch uploading.

- **Endpoint:** POST /collection/sync
- **Access:** Requires TA Role

Request Body:

```JSON
{
  "collections": [
    {
      "supplier_id": "c9d8e7f6-a5b4-3210-fedc-0987654321fe",
      "gross_weight": 45.50,
      "gps_lat": 6.8415,
      "gps_long": 79.9018,
      "collection_time": "2026-02-21T07:15:00Z"
    },
    {
      "supplier_id": "d1e2f3a4-b5c6-7890-abcd-112233445566",
      "gross_weight": 30.00,
      "gps_lat": 6.8450,
      "gps_long": 79.9050,
      "collection_time": "2026-02-21T08:00:00Z"
    }
  ]
}
```

Response (201 Created):

```JSON
{
  "message": "Collections synced successfully",
  "synced_records": 2,
  "failed_records": 0
}
```

3.2 Get Leaf Supply History

Retrieves the brought leaf history for a specific supplier.

- **Endpoint:** GET /collection/history/{supplierId}
- **Access:** Requires SMALL_HOLDER, EXT, MG, or TA Role
- **Query Parameters:** ?startDate=2026-02-01&endDate=2026-02-21

Response (200 OK):

```JSON
{
  "supplier_id": "c9d8e7f6-a5b4-3210-fedc-0987654321fe",
  "total_net_weight": 75.50,
  "history": [
    {
      "collection_id": "f1a2b3c4-...",
      "date": "2026-02-21T07:15:00Z",
      "gross_weight": 45.50,
      "water_deduction": 2.00,
      "net_weight": 43.50,
      "quality_grade": "A"
    }
  ]
}
```

4. Financial Ledger Endpoints

4.1 Request Financial Advance

Allows a Small Holder (or a TA on their behalf) to request an advance payment.

- **Endpoint:** POST /finance/advance-request
- **Access:** Requires SMALL_HOLDER or TA Role

Request Body:

```JSON
{
  "supplier_id": "c9d8e7f6-a5b4-3210-fedc-0987654321fe",
  "requested_amount": 5000.00,
  "reason": "Emergency agricultural supplies"
}
```

Response (201 Created):

```JSON
{
  "message": "Advance request submitted and pending approval.",
  "transaction_id": "b2c3d4e5-...",
  "status": "PENDING"
}
```

4.2 Get Financial Ledger Details

Fetches the current financial standing (debts, advances, payouts) of a Small Holder.

- **Endpoint:** GET /finance/ledger/{supplierId}
- **Access:** Requires SMALL_HOLDER, EXT, MG, or SK Role

Response (200 OK):

JSON

```
{
  "supplier_id": "c9d8e7f6-a5b4-3210-fedc-0987654321fe",
  "summary": {
    "total_advances": 5000.00,
    "total_debts": 1200.00,
    "pending_payout": 15500.00
  },
  "transactions": [
    {
      "transaction_id": "...",
      "type": "DEBT",
      "amount": 1200.00,
      "description": "Fertilizer Cost (Urea 50kg)",
      "date": "2026-02-15T10:00:00Z",
      "status": "APPROVED"
    }
  ]
}
```

5. Service & Inventory Endpoints

5.1 Create Service Request

Submit a request for fertilizer, machinery, or transport.

- **Endpoint:** POST /services/request
- **Access:** Requires SMALL_HOLDER or TA Role

Request Body:

```JSON
{
  "supplier_id": "c9d8e7f6-a5b4-3210-fedc-0987654321fe",
  "request_type": "FERTILIZER",
  "item_id": "e8d9c0b1-...",
  "quantity": 2
}
```

Response (201 Created):

```JSON
{
  "message": "Service request created.",
  "request_id": "a1b2c3d4-...",
  "status": "PENDING"
}
```

5.2 Update Service Request Status (Approval)

Used by Management or Store Keepers to approve/dispatch requested items.

- **Endpoint:** PATCH /services/request/{requestId}/status
- **Access:** Requires MG, EXT, or SK Role

Request Body:

```JSON
{
  "status": "DISPATCHED",
  "approved_by_id": "a1b2c3d4-e5f6-7890-abcd-1234567890ab"
}
```

Response (200 OK):

```JSON
{
  "message": "Request status updated successfully.",
  "request_id": "a1b2c3d4-...",
  "new_status": "DISPATCHED"
}
```